



OPERATIONAL MANUAL

SMART+ PASS BOX CONTROLLER – 2 DOOR

MODEL NO: SMART+ 2D

VERSION: 1.0



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Dear Customer,

Thank you for choosing a NEURAL instrument, please read this operating manual carefully before you start using the instrument. This manual will provide you with all the necessary information for the correct use of the instrument as well as a precise idea of its versatility in a wide range of applications.

CAUTION

One three way connector is provided for power connection i.e. Line, Neutral & Earth. (100-220V AC 50/60Hz.)



Do not Switch ON power to the instrument before plugging connectors in to the terminals.



Carefully connect the connectors in the specified terminals if not it may cause damage for the instrument.

Instrument SI. No :**STANDARD PACK**

1) Instruction Manual	---	1 No	☐
2) SMART+ 2D Instrument	---	1 No	☐

Checked By :

Date :

1. Description:

Pass box is used for material transfer from one Clean room to another clean room. It consists of a rectangular box with two doors. The two doors are interlocked such that two doors cannot be open at the same time, to prevent pressure losses and cross contamination.

Our Pass Box controller achieve this by preventing simultaneous access of door by ensuring that when one door for a Pass box is open then other door is closed.

Before opening of door,

1. When Door 1 is opened the display on the Door 2 will indicate “WAIT ”
2. When any of the Door is kept open beyond a particular time say 2 min, a buzzer will be blown
3. When any door is opened the CF lamp will be ON automatically and also CFL ON/OFF by using soft switch at any time.
4. When both doors are closed the UV lamp will be ON automatically and also UVL ON/OFF by using soft switch when both doors are closed.
5. Both Doors are provided with EM Locks
6. Blower fan will be ON/OFF by using soft switch at any time

2. Features:

- Advanced Micro controller based.
- Supports Maximum of 2 doors
- User settable door delay time from 0 – 250sec/ min.
- Supports Electromagnetic locks/Electric strikers
- Inbuilt UV hour meter
- Door status indication with high intensity Bar LEDs
- Status indication: GO, WAIT
- Blower MCB trip indication with alarm
- Buzzer output for door is kept open more than 2 min.

- UV Lamp controlling
 - UV lamp ON/OFF using soft switch.
- CF Lamp controlling
 - CF lamp ON/OFF using soft switch.
- Blower controlling
 - Blower ON/OFF using soft switch.

3. Specifications:

GENERAL

- Electro Magnetic Lock Source : 24V DC/ 250 mA

Door Control by Relay:

RELAYS:

- Contact material : Silver Alloy.
- Contact Rating : 230V AC / 2 A.
- No. of Operations : Mechanical : 10^7 , Electrical : 10^5 .

ENCLOSURE:

Pass Box Controller

- Enclosure type : Power coating Box
- Mounting : Wall

Display Module

- Enclosure type : SS-316L
- Mounting : Panel

POWER SUPPLY:

- Power : 80 - 280V, AC 50 Hz
- Wattage : 80W

ENVIRONMENTAL:

- Operating Temperature : 0 to 50°C.

4. Electromagnetic locks:

A **magnetic lock** is a simple locking device that consists of an electromagnet and armature plate. By attaching the electromagnet to the door frame and the armature plate to the door, a current passing through the electromagnet attracts the armature plate holding the door shut. The magnetic lock relies upon some of the basic concepts of electromagnetism. Essentially it consists of an electromagnet attracting a conductor with a force sufficiently large enough to prevent the door from being opened.



Features:

- Operating voltage 24V
- 300/600 lbs holding force
- Built in Magnetic Contact for Door Status Monitoring

5. Status Indication on Display Panel:

GO - Indicates that person can access the door

WAIT - indicates that person cannot access the door

UV LAMP:

Green LED = Indicate that UV lamp is on.

Red LED = Indicate that UV lamp is off.

BLOWER:

Green LED = Indicate that BLOWER is on.

Red LED = Indicate that BLOWER is off.

CF LAMP:

Green LED = Indicate that CF lamp is on.

Red LED = Indicate that CF lamp is off.



6. Status Indication on Push button (optional):

GO = Indicates that person can access the door.

WAIT= Indicate that person can't access the door



7. OPERATING MODES:

Normally Close:

In this mode, initially both doors in locked condition. Door access by using push button, Status LED's provided on the Display module indicates whether person can access the door or not.

GO - indicates that person can access the door, **WAIT** - indicates that person cannot access the door.

When the door operated from open to closed condition, both doors locked for default 20Sec., with in this time person cannot access both doors. This time can be settable from 0-250sec/min. Auto close time delay i.e. the time between the door release and energizing is 4 Sec.

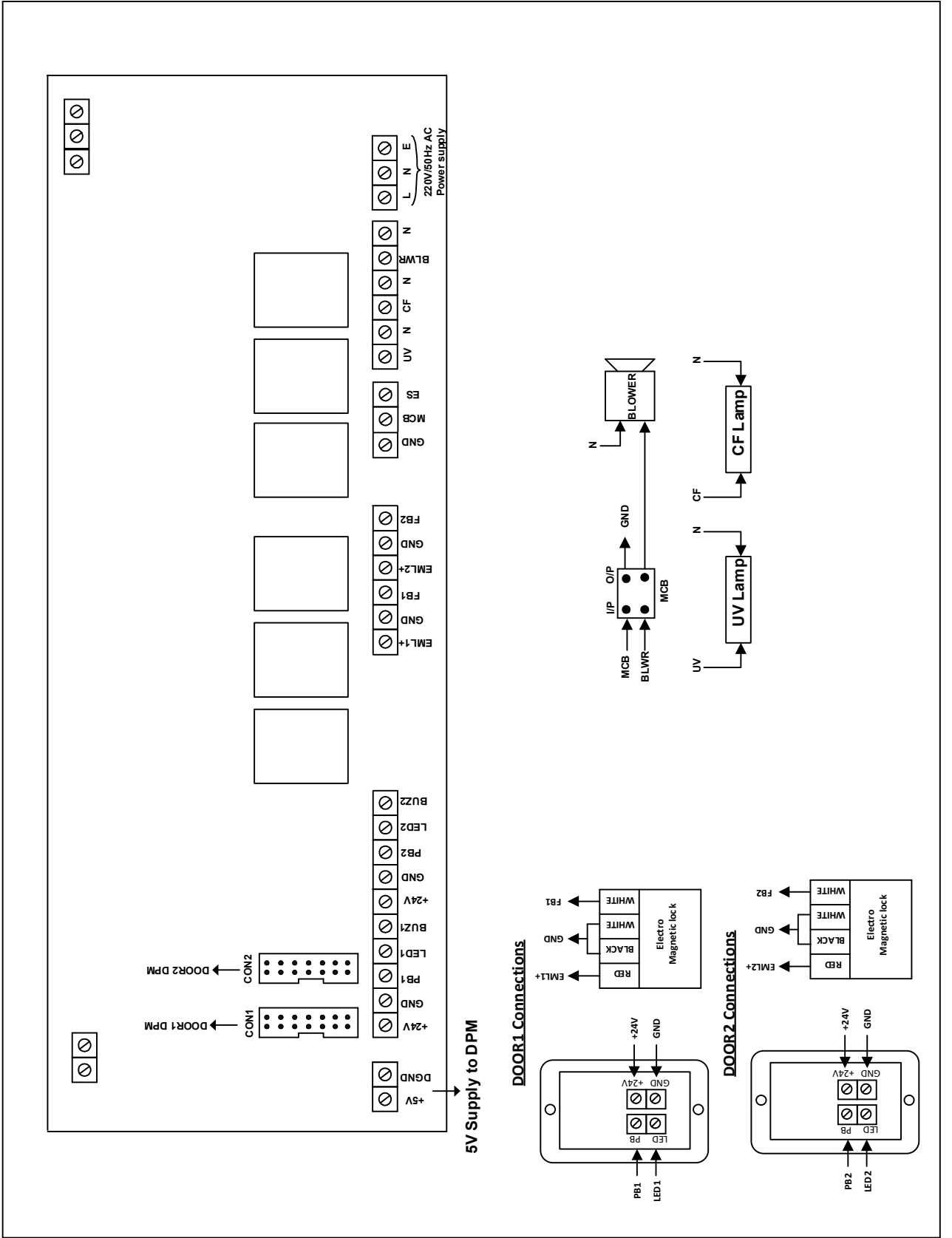
8. Programming Mode:

Door Delay Time setting:

S. No	Action to perform	Action to be Takes placed	Display
1.	To enter into configuration mode.	Press ▲/▼ key, then it will ask for password entry.	P5d
	Edit the password to 005	Use ▲/▼ to edit the value and press OK to conform.	005
2.	Then display shows Door1 delay time		d1t
		Use ▲/▼ to edit the Door1 time and press OK to conform.	020
3.	Then display shows Door2 delay time		d2t
		Use ▲/▼ to edit the Door2 time and press OK to conform.	030

4.	Then display shows SEC/MIN mode selection		SEC
		Use ▲/▼ to edit SEC/MIN mode and press OK to conform.	Min
5.	Then display shows new password selection.		P5n
		Use ▲/▼ to edit the value and press OK to conform.	007

9. Wiring diagram:



10. MECHANICAL DRAWINGS

Fig (10.1) Figure for Installation of SMART+ 2D Controller:

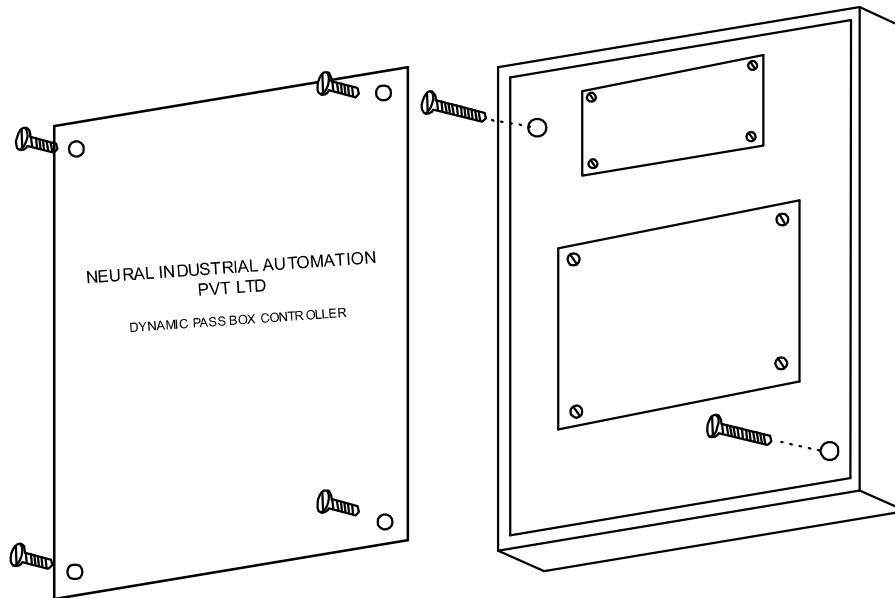
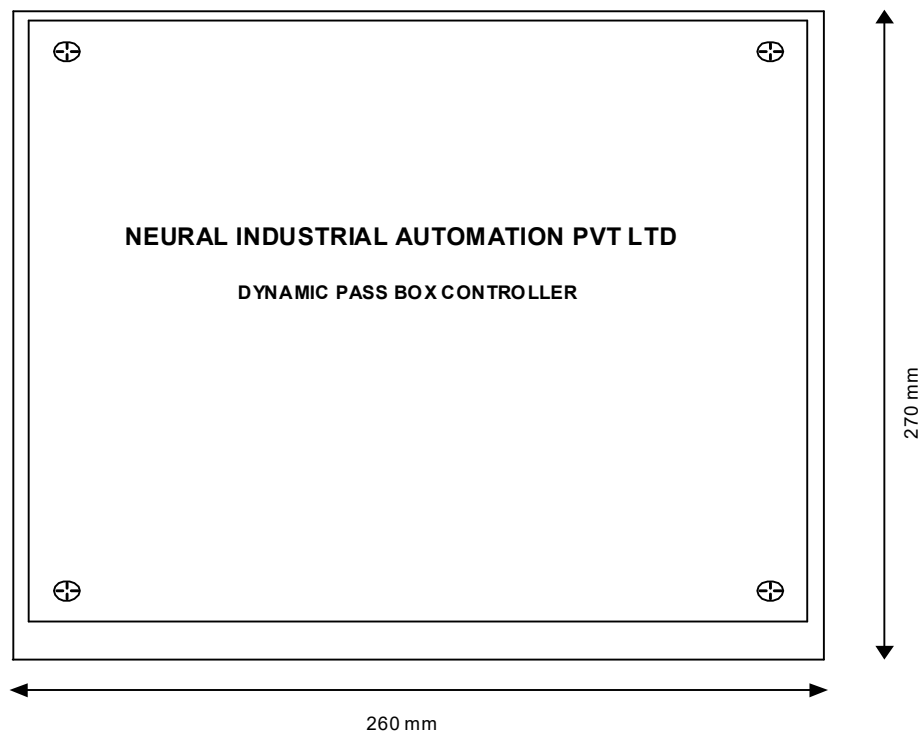


Fig (10.2) Front View Diagram of SMART+ 2D Controller:



11. Ordering Information:

I). Pass Box Controller

	X
<i>Model No</i>	<i>Pass Box Type</i>
SMART+	2D- 2DOOR
	3D- 3DOOR

includes,

a. Controller:

	X
<i>Model No</i>	<i>Pass Box Type</i>
SMART+C	2D- 2DOOR
	3D- 3DOOR

Variants:

SMT153	X	Model. No.
SMART+C	2D	SMART+ -2D
SMART+C	3D	SMART+ -3D

b. Pass Box Display Module:

SMART+DPM	X
<i>Model No</i>	<i>Future Use</i>
SMART+DPM	X

NOTE:

For Further Information Contact:

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